REAL-TIME KUBERNETES MICROSERVICES

Production DevOps Project — Food Delivery Platform

A hands-on guide for React, FastAPI microservices, Docker, Kubernetes, Helm, Terraform, Azure AKS, CI/CD, security and observability.

1. Project Overview

React frontend, API Gateway, User, Restaurant, Order, Payment and Notification services. PostgreSQL provides durable data; Redis provides caching and temporary order state.

2. Runtime Architecture

User → Ingress → Frontend/API Gateway → microservices → PostgreSQL/Redis. Kubernetes Services provide internal discovery. Probes, replicas, rolling updates, HPA, PDB and NetworkPolicy support production-style operations.

3. Local Docker

Run `docker compose build`, `docker compose up -d`, `docker compose ps`, then open port 8080. Use `docker compose logs -f` for troubleshooting and `docker compose down` for cleanup.

4. Docker Security

Application images use slim/alpine bases, non-root execution where practical, dropped capabilities, health checks and .dockerignore files. Scan images with Trivy before pushing.

5. Kubernetes

Apply namespace, ConfigMap, Secret, RBAC, database/cache, service deployments, HPA, Ingress and NetworkPolicy. Each application has startup, readiness and liveness probes plus resource requests/limits.

6. Helm

Use `helm lint`, `helm template`, `helm upgrade --install`, `helm rollback` and `helm uninstall`. Values control registry, image tag, replicas, database and ingress settings.

7. Terraform + AKS

Terraform provisions resource group, VNet/subnet, AKS, ACR, Log Analytics and the AKS kubelet AcrPull assignment. Review region, VM size, scaling, networking and security before production.

8. CI/CD

Git push → validation/tests → Docker build → Trivy scan → registry push → AKS image update → rollout status → smoke checks. Use workload identity/OIDC or approved federated credentials rather than long-lived secrets.

9. Security

Never commit real credentials. Use an external secret manager such as Azure Key Vault with CSI or another approved platform. Apply least-privilege RBAC and NetworkPolicy. Prefer immutable image tags or digests in production.

10. Monitoring & Logging

Prometheus/Grafana should track CPU, memory, restarts, API rate/latency, HPA and database health. Use kubectl logs, logs --previous and describe pod during incidents. Centralized logging should aggregate structured application logs.

11. Troubleshooting

CrashLoopBackOff: inspect current/previous logs and environment. ImagePullBackOff: inspect pod events and registry permissions. Pending: inspect scheduling/PVC events. OOMKilled: inspect metrics and tune limits. Probe failure: test endpoint and startup window. Ingress/service/DNS failures: inspect endpoints, controller logs and NetworkPolicy.

12. Disaster Recovery

Take PostgreSQL logical backups, store them in durable encrypted storage, test restores, retain immutable container images, and keep deployment rollback procedures documented. Define RPO/RTO with stakeholders.

13. AWS/GCP Equivalents

For AWS, use EKS + ECR and cloud-native IAM/load balancing/storage. For GCP, use GKE + Artifact Registry and Workload Identity. Kubernetes application manifests remain largely portable.

14. Interview Preparation

Be ready to explain API Gateway, service discovery, probes, HPA/PDB, rolling updates, NetworkPolicy, immutable tags, registry security, CI/CD gates, database resilience and observability.

Useful Commands

```
docker compose build
docker compose up -d
docker compose ps
docker compose logs -f

terraform init
terraform validate
terraform plan
terraform apply

kubectl -n food-delivery get pods,svc,ingress,hpa
kubectl -n food-delivery logs -f deployment/order-service
kubectl -n food-delivery describe pod <pod>
kubectl -n food-delivery rollout status deployment/order-service
kubectl -n food-delivery rollout undo deployment/order-service

helm lint helm/food-delivery
helm template food-delivery helm/food-delivery
helm upgrade --install food-delivery helm/food-delivery -n food-delivery --create-namespace

trivy image --severity HIGH,CRITICAL --ignore-unfixed <image>
```